

Balaji Bheemavarapu

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PROFILE

Full Stack Software Engineer with **4+ years of experience** building enterprise applications using React, Next.js, Node.js, TypeScript, REST APIs, micro-frontends and microservices. Experienced in **secure development, production troubleshooting, performance optimization, API integration, testing, and CI/CD**, with exposure to **Azure**. Skilled in **LLM integration and AI-powered applications**, collaborating with **engineering, product and business stakeholders** in Agile/Scrum teams.

PROFESSIONAL EXPERIENCE

Full Stack Engineer, TDCX Digilab India Pvt Ltd

02/2026 – 09/2026

Summary: Contributed to an AI-powered enterprise Agent Assist platform, developing customer-support workflows and browser-extension features across frontend and backend using modern React/Node.js technologies.

- Developed enterprise features using **React, Next.js, Node.js and TypeScript**, contributing across frontend, backend API integration, testing and production support.
- Built and enhanced **React/TypeScript modules and Chrome Extension workflows**, focusing on responsive UI, asynchronous data flows and application performance.
- Implemented **REST API integrations and asynchronous workflows**, including Redis-based caching and AI-enabled functionality.
- Gained **hands-on exposure to GCP** while working with cloud-based application workflows and development environments.
- Collaborated with cross-functional engineering teams on feature development, debugging, code quality and production issue resolution.

Tech stack: React, Next.js, Node.js, TypeScript, Plasmo, REST APIs, Redis, GCP, Claude via VS code.

Full Stack Developer, Tata Consultancy Services

06/2023 – 01/2026

Summary: Modernized high-traffic UK banking and mortgage applications, contributing across frontend, backend APIs, micro-frontends, security, performance, testing and production deployment.

- Developed and modernized **high-traffic UK-based banking applications** using **React, TypeScript, Node.js, micro-frontends and microservices**, supporting critical mortgage and customer-facing workflows.
- Designed and integrated **REST APIs with Node.js**, with API gateway integration through **Azure API Management**, contributing to approximately **35% reduction in API latency**.
- Automated **production deployment activities**, reducing manual deployment effort by **~90%** and improving release efficiency.
- Implemented **caching, lazy loading and parallel API/data fetching** to improve application performance and responsiveness, reducing page load time from approximately **3 seconds to milliseconds** for key workflows.
- Worked across **frontend and backend layers**, including feature development, API integration, debugging, production issue resolution and performance optimization within an enterprise banking environment.
- Developed **unit and integration test cases using Jest and React Testing Library**, contributing to **95%+ test coverage** and improving release confidence.
- Performed and supported **Azure production deployments through Azure DevOps**, including release validation, deployment checks and production troubleshooting.
- During a major **platform transformation**, took **Scrum facilitation responsibilities** on a rotating basis, coordinating daily stand-ups, tracking blockers and supporting sprint execution; also owned **ticket refinement, requirement clarification, task breakdown and delivery planning** in collaboration with the **ETM, Product Owner and business stakeholders**.

- Actively participated in **sprint planning, backlog refinement, retrospectives, reviews and technical discussions**, collaborating with an **8-member Agile/Scrum engineering team**.
- Used **Dynatrace** for **production diagnostics, troubleshooting and application/API performance analysis**.

Tech stack: React 18+, TypeScript, Node.js, Micro-frontends, Microservices, REST APIs, Azure API Management, Azure DevOps, JWT, RBAC, Jest, React TestingLibrary, Dynatrace.

Software Engineer Trainee (Intern), Infolabs Private Ltd

06/2022 – 05/2023

Summary: Developed an enterprise Video-on-Demand platform with responsive, multi-device experiences and no-code content customization capabilities

- Developed a **Video-on-Demand (OTT) platform** using React, Redux, TailwindCSS, Material UI and GraphQL for multi-device playback and adaptive UI rendering.
- Built no-code customization modules using a GraphQL CMS, enabling non-technical users to configure OTT themes and reducing developer support effort by **40%**.
- Collaborated with backend & design teams; followed Agile & GitLab workflows.

Tech stack: React.js, Redux, TailwindCSS, Material UI, GraphQL, GraphQL CMS, GitLab

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python

Frontend: React, Next.js, Redux, Micro Frontends, Tailwind CSS, Material UI, Bootstrap

Backend: Node.js, Express.js, REST APIs, Microservices, Redis, MongoDB, MySQL

Cloud & DevOps: Azure, Azure API Management, Docker, Azure DevOps, Webpack, GitHub Actions, CI/CD

Security & Testing: OAuth 2.0, JWT, Authentication, Authorization, RBAC, Jest, React Testing Library, Dynatrace, SonarQube

AI/GenAI: LLM Integration, GPT-4, OpenAI API, Gemini, Claude, Prompt Engineering

PERSONAL PROJECTS

AI Agent for Automated User Story Breakdown, React, Node.js, Gemini Pro, Docker

- **Led 5-member** team to build an AI-driven user story-to-task breakdown solution within a 6-hour hackathon.
- Developed a production-ready prototype using **React, TailwindCSS, Node.js, Express, and Docker, integrating Gemini 2.5** (migrated from Azure GPT-4.0) for intelligent task generation.

Automated Programming Evaluation, apecode.bbalajis.com

- Developed a **real-time code evaluator** for C, C++, Java & Python with role-based access and test-case scoring.
- Deployed via **GitHub Actions** (CI/CD) and tested with **60+ active users**, featuring a responsive **Tailwind + MUI** interface built with **modular Micro UI design principles**

ACHIEVEMENTS

Certificate of Excellence - Lloyds Banking Group CIO for contribution to mortgage process automation.

2nd Prize - TCS AI Fridays Hackathon for developing an AI Agent for User Story Breakdown.

TCS Top Coders - Top 3% among 7,000+ developers.

TCS CodeVita Season 10 - Global Rank 3604 among 100K+ participants.

EDUCATION

Bachelor of Technology, GMRIT

2019 – 2023

Department: Information Technology CGPA: 8.22